Spectral–Temporal Consistency Prior for Cloud Removal From Remote Sensing Images

Shi-Jun Yang, Yu-Bang Zheng, *Member, IEEE*, Heng-Chao Li, *Senior Member, IEEE*, Yong Chen, and Qing Zhu

***Abstract*— Thick cloud removal for multitemporal remote sensing images (MTRSIs) is a necessary preprocessing step for subsequent applications. Existing methods for cloud removal ignore spectral–temporal consistency prior (STCP), such as smooth regions of different time and bands existing in the same spatial location. To address this problem, we propose a factor-based group sparsity regularization within the low-rank tensor factorization (LRTF) framework and theoretically prove that it can characterize the STCP in MTRSIs. Based on this regularization, we construct a cloud removal model for MTRSIs. On one hand, the introduction of STCP enables the model to achieve superior cloud removal performance. On the other hand, regularization on small-sized factors rather than on the original data enables the model to have extremely low computational complexity. To solve this model, we develop a proximal alternating minimization (PAM)-based algorithm, in which we integrate a mask acquisition method based on separated cloud and shadow components. Comparative experiments using both simulated and real data demonstrate that the proposed method outperforms recent mask-unknown and mask-known methods in terms of performance and efficiency.**

***Index Terms*— Cloud removal, group sparsity, multitemporal remote sensing images (MTRSIs), proximal alternating minimization (PAM), spectral–temporal consistency prior (STCP).**

NOMENCLATURE

$x, \mathbf{x}, \mathbf{X}, \mathcal{X}$	Scalar, vector, matrix, tensor.
$\mathcal{X}_{(i_1,i_2,:,i_4)}$	(i_1, i_2, i_4)th tube of a fourth-order tensor $\mathcal{X}$.
$\mathcal{X}_{(i,j,:)}$	(i, j)th tube of a third-order tensor $\mathcal{X}$.
$x_{i_1,i_2,\ldots,i_K}$ or $\mathcal{X}_{(i_1,i_2,\ldots,i_K)}$	$(i_1, i_2, \ldots, i_K)$th element of a Kth-order tensor $\mathcal{X}$.
$\mathbf{X}_{(k)}$	Mode-k unfolding of a tensor $\mathcal{X}$.
$\|\mathcal{X}\|_{2,1}$	$\ell_{2,1}$-norm of third-order tensors $\|\mathcal{X}\|_{2,1} = \sum_{i_1,i_2} \|\mathcal{X}_{(i_1,i_2,:)}\|_2$.
$\|\mathcal{X}\|_F$	Frobenius norm of the Kth-order tensors $\|\mathcal{X}\|_F = \sqrt{\sum_{i_1,i_2,\ldots,i_K} \|x_{i_1,i_2,\ldots,i_K}\|^2}$.
$\times_k$	Mode-k tensor-matrix product.
$\odot$	Componentwise multiplication.
$\otimes$	Kronecker product.
$\oslash$	Componentwise division.

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Shi-Jun Yang, Yu-Bang Zheng, and Heng-Chao Li are with the School of Information Science and Technology, Southwest Jiaotong University, Chengdu 611756, China (e-mail: 18782927798@163.com; zhengyubang@163.com; hcli@home.swjtu.edu.cn).

Yong Chen is with the School of Computer and Information Engineering, Jiangxi Normal University, Nanchang 330022, China (e-mail: chenyong1872008@163.com).

Qing Zhu is with the Faculty of Geosciences and Engineering, Southwest Jiaotong University, Chengdu 611756, China (e-mail: zhuq66@263.net).

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I. INTRODUCTION

THE existing satellite technology and equipment enable us to acquire high spatial resolution multitemporal remote sensing images (MTRSIs), which are used for various applications such as land-use classification [1], [2], field environmental monitoring [3], [4], [5], and urban investigation [6], [7], [8]. Unfortunately, cloudy weather can lead to a significant absence of information for specific temporal segments of MTRSIs. The cloud not only contaminates a large spatial area but also often affects all the bands, which impedes the subsequent progression such as super-resolution and unmixing [9]. Therefore, the reconstruction of segments contaminated by cloud and shadow in MTRSIs is a necessary and challenging undertaking.

A. Related Works

Many methods have now been devoted to cloud removal for MTRSIs. Depending on whether a given cloud/shadow mask is required, cloud removal methods can be classified as mask-known methods and mask-unknown methods. Mask-known methods rely on given cloud/shadow masks to locate and reconstruct the contaminated areas, which can be divided into the following categories: temporal information-based, spatial information-based, spectral information-based, and hybrid methods.

1) *Temporal information-based methods:* They exploit the consistency and similarity of details in MTRSIs captured at different time nodes in the same area. Methods [10], [11], [12] use information cloning to replace the contaminated regions in the images captured at the contaminated time nodes with the uncontaminated images from other time nodes. However, these methods have difficulty in accurately improving the features of the uncontaminated images used for replacement to match the target images.

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2) *Spatial information-based methods:* They are also widely used. Guillemot and Le Meur [13] argued that neighboring areas affected and unaffected by cloud contamination contain similar information. Interpolation methods [14] effectively used spatial information from the uncontaminated areas. Methods based on various variations [15] and exemplars [16] also achieved good cloud removal results using only spatial information. However, due to the widespread distribution of clouds at certain time nodes, the available spatial information is reduced. Therefore, these methods are only suitable for cases where the contaminated areas are relatively small and have few texture details [17]. Otherwise, the accuracy of reconstructing the contaminated regions will be severely limited.
3) *Spectral information-based methods:* They mainly use the similarities and correlations between different spectral bands. Lv et al. [18] used additional spectral information to address the loss of spectral information caused by sensor failures or thin clouds. Xu et al. [19] considered both the spatial and multispectral domains in cloud signal-to-noise ratio estimation. However, in real scenarios, thick clouds often contaminate all the bands. Spectral-based methods cannot obtain sufficient reference information, so using only spectral and spatial information still cannot achieve effective cloud removal.
4) *Hybrid methods:* They take the spatial, temporal, and spectral information of MTRSIs into consideration [20]. Tensor completion and sparse representation can make good use of these multidimensional information for various image processing tasks [21], [22], [23], [24], [25].

For example, Zhu et al. [26] proposed a low-rank tensor completion video smoke elimination method based on spatial–temporal continuity constraints, which is of enlightening significance. Xu et al. [27] adopted fractal componentwise modeling based on group sparsity on the image fusion task. Deng et al. [28] proposed a model combining group sparsity residual and low-rankness for image denoising. Some methods [29] treat MTRSIs cloud removal as a tensor completion task using tensor decomposition [30], [31] such as tensor singular value decomposition (t-SVD) [32], tensor train (TT) decomposition [33], tensor ring (TR) decomposition [34], and tensor network decomposition [35]. Zhang and Aeron [36] proposed a t-SVD-based tensor completion method for cloud removal. He et al. [37] proposed a remote sensing image reconstruction method based on TR completion and total variation (TV). Li et al. [38] combined TR decomposition with gradient domain fidelity to build a cloud removal model. Since mask-known methods can obtain additional accurate information about the contaminated regions from cloud/shadow masks, they often achieve eligible cloud removal results. However, not all the MTRSIs have corresponding cloud/shadow masks. In addition, due to the similarity between cloud edge pixel values and certain high-light areas in the image, as well as the similarity between shadow pixel values and darker areas in the image, obtaining complete and accurate cloud/shadow masks is extremely challenging. The use of inaccurate cloud masks will significantly compromise the cloud removal results of mask-known methods.

Mask-unknown methods overcome the problem of heavy reliance on cloud/shadow masks. Duan et al. [39] proposed a remote sensing image thick cloud removal method based on temporal smoothness and sparsity regularization tensor optimization (TSSTO). The method proposed by Xu et al. [40] separately learned dictionaries for cloudy and cloud-free regions in the spectral domain, which effectively restored data contaminated by thin and thick clouds or shadows. With the improvement of computility, deep learning was widely applied in remote sensing image processing [41], [42], [43]. For example, Zhang et al. [44] proposed a lightweight transformer-based cloud detection network. Chen et al. [45] used deep networks under only block-level supervision to achieve pixel-level cloud detection. Li et al. [46] proposed a weakly supervised deep-learning-based cloud detection, which effectively reduces the amount of labor required for pixel-level label annotation. Moreover, they presented a novel hierarchical spectral and structure-preserving fusion network [47] based on the hierarchical fusion of optical and SAR remote sensing images to reconstruct cloud and shadow regions. Meraner et al. [48] designed a deep residual neural network architecture using SAR-optical data fusion techniques and the collaborative characteristics of two imaging systems for cloud removal from multispectral Sentinel-2 images. Zhang et al. [49] used the MSCFF method [50] to obtain cloud masks and combined deep prior with low-rank t-SVD. Semi-masked methods with partial knowledge of cloud positions can also accomplish cloud removal tasks without accurate masks. For instance, Lin et al. [51] described spectral and temporal correlations using coupled tensor factorization in the cloud removal process but ignored the correlation between two spatial patterns. To address this issue, Zheng et al. [52] proposed a spatial–spectral–temporal connective tensor network decomposition method for cloud removal in MTRSIs.

B. Motivation

The above methods show good performance in cloud removal from MTRSIs. However, they still have two problems. First, the existing methods overlook the spectral–temporal consistency prior (STCP), such as smooth regions of different time and bands existing in the same spatial location. This prior makes the spatial difference images in two spatial directions of MTRSIs and the separated cloud components exhibit a large number of columns that are either equal to zero or close to zero when unfolded along the third dimension [as shown in Fig. 1(b) and 1(c)]. Since the contaminated images lack spectral–temporal consistency, the process of promoting the spectral–temporal consistency is also the process of cloud removal. Second, the mask acquisition step and the cloud removal model in the existing methods are independent of each other, resulting in the cloud removal model unable to contribute to the detection for obtaining a more precise mask.

To address the first issue, we propose group sparsity regularization on both the spatial factor and cloud component within the low-rank tensor factorization (LRTF) framework. This regularization precisely captures the STCP in MTRSIs. To tackle the second issue, we use the proposed method

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to separate cloud and shadow components and construct a thresholding algorithm to acquire an accurate mask. This approach can accurately encompass all the cloud and shadow regions with minimal false detection. By integrating the aforementioned group sparse regularization and mask acquisition techniques, we construct a fast cloud removal model for MTRSIs. In summary, this article makes the following three main contributions.

1) Within the LRTF framework, we propose $l_{2,1}$-norm-based group sparsity regularization on spatial difference images of spatial factor and cloud component, which can effectively characterize STCP. We prove in Theorem 2 that regularization on spatial factor can replace regularization on original images with same effect, but only requires much less computational complexity.
2) Based on the aforementioned group sparsity regularization, we develop a fast cloud removal model for MTRSIs. This model can characterize not only the global low-rank prior but also the STCP, that is, smooth regions of different time and bands existing in the same spatial location.
3) We develop an algorithm based on proximal alternating minimization (PAM) to solve the proposed model,[1] incorporating a threshold-based algorithm using the cloud and shadow components to obtain accurate cloud and shadow masks. The proposed method exhibits superior performance and computational efficiency compared with recent advanced mask-unknown and mask-known methods.

The main contents of the remaining sections in this article are outlined as follows. The main notations and preliminaries of tensors covered in this article are presented in Section II. Section III introduces how the proposed weighted group sparsity regularized LRTF cloud removal model characterizes the STCP and optimization procedure. Section IV reports experiments on both the simulated and real data to illustrate the superiority of the proposed model, the impact of parameters, and convergence analysis. Section V concludes this article.

II. Notations and Preliminaries

In this section, we show in Nomenclature the interpretations of notations in this article. In addition, we introduce the definitions of mode-k unfolding, mode-k product based on [56], and Sylvester matrix equation [57].

Definition 1 (Mode-k Unfolding): Let $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_K}$ be a Kth-order tensor. Its mode-k unfolding result is an $I_k \times \prod_{n=1,n\neq k}^{K} I_n$ matrix $\mathbf{X}_{(k)}$, whose element $(\mathbf{X}_{(k)})_{(i_n,j)}$ is mapped by $\mathcal{X}_{(i_1,i_2,\ldots,i_K)}$, where $j = 1 + \sum_{n=1,n\neq k}^{K}(i_n - 1)\prod_{m=1,m\neq k}^{n-1} I_m$. Hereinafter, we denote the mode-k unfolding operator by $\text{Unfold}_k(\mathcal{X})$ and denote the inverse operator by $\text{Fold}_k(\mathbf{X}_{(k)})$.

Definition 2 (Mode-k Product): Let $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_K}$ be a Kth-order tensor and $\mathbf{E} \in \mathbb{R}^{J \times I_k}$ be a matrix. Their mode-k product is a tensor denoted by $(\mathcal{X} \times_k \mathbf{E}) \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_{k-1} \times J \times I_{k+1} \times \cdots \times I_K}$ which satisfies $(\mathcal{X} \times_k \mathbf{E})_{(i_1,i_2,\ldots,i_{k-1},j,i_{k+1},\ldots,i_K)} = \sum_{i_k=1}^{I_k} x_{i_1,i_2,\ldots,i_K} a_{j,i_k}$.

[1]The proposed model can also be solved using the Bregman algorithm [53], half-quadratic splitting (HQS) [54], and alternating direction method of multipliers (ADMM) [55]. However, these methods will lead to more complex computations.

Theorem 1 (Fast Solution of Sylvester Matrix Equation): Suppose that $\mathbf{M} \in \mathbb{R}^{m\times m}$, $\mathbf{N} \in \mathbb{R}^{n\times n}$, and $\mathbf{X}$, $\mathbf{Y} \in \mathbb{R}^{m\times n}$. If $\mathbf{I}_n \otimes \mathbf{M} + \mathbf{N}^{\mathsf{T}} \otimes \mathbf{I}_m$ is an invertible matrix, the classical Sylvester matrix equation

$$\mathbf{MX} + \mathbf{XN} = \mathbf{Y}$$

has a unique solution. In addition, if $\mathbf{M}$ and $\mathbf{N}$ can be decomposed into

$$\mathbf{M} = \mathbf{F}\Psi\mathbf{F}^{\mathsf{T}} \text{ and } \mathbf{N} = \mathbf{U}\Sigma\mathbf{U}^{\mathsf{T}}$$

we can get a unique solution to $\mathbf{X}$

$$\mathbf{X} = \mathbf{F}\big((1 \oslash \mathbf{T}) \odot (\mathbf{F}^{\mathsf{T}}\mathbf{Y}\mathbf{U})\big)\mathbf{U}^{\mathsf{T}}$$

where Ψ and Σ are the diagonal matrices, $\mathbf{F}$ and $\mathbf{U}$ are the unitary matrices. And assuming that $\text{diag}(\Psi)$ denotes column vector whose elements are diagonal elements of Ψ, then $\mathbf{T}$ is defined as

$$\begin{aligned}\mathbf{T} = &(\text{diag}(\Psi), \text{diag}(\Psi), \ldots, \text{diag}(\Psi)) \\ &+ (\text{diag}(\Sigma), \text{diag}(\Sigma), \ldots, \text{diag}(\Sigma))^{\mathsf{T}}.\end{aligned}$$

III. Proposed Cloud Removal Model and Solving Algorithm

Let $\mathcal{Y} \in \mathbb{R}^{n_1\times n_2\times n_3\times n_4}$ be the observed cloud/shadow-contaminated MTRSIs data, where n_1 and n_2 are the spatial height and width, respectively; n_3 is the number of spectral bands, and n_4 is the number of time nodes. The degradation model can be written as follows:

$$\mathcal{Y} = \mathcal{X} + \mathcal{S} + \mathcal{N} \tag{1}$$

where $\mathcal{X}$ represents the clean MTRSIs' component, $\mathcal{S}$ represents the cloud component, and $\mathcal{N}$ represents the residual. The goal of our work is to make the clean components estimated by the model approximate the true values separated from the observed data as closely as possible.

A. Proposed Model

To accurately estimate the separated clean and cloud components of MTRSIs, it is important to leverage their inherent properties effectively. MTRSIs have a limited number of spectral bands at a single time node, and the contamination of any band by clouds implies contamination of all the bands. Therefore, the uncontaminated spectral information in an individual image captured at a specific time node is highly limited. To supplement the insufficient spectral information, we stack the images $\mathcal{X}_i \in \mathbb{R}^{n_1\times n_2\times n_3}$ $(i = 1, 2, \ldots, n_4)$ from different time nodes along the spectral dimension, creating a third-order tensor $\mathcal{R}(\mathcal{X})$ using the reshape operator which is denoted by $\mathcal{R}(\cdot) : \mathbb{R}^{n_1\times n_2\times n_3\times n_4} \to \mathbb{R}^{n_1\times n_2\times n_3 n_4}$.

For stacked clean component $\mathcal{R}(\mathcal{X})$, its adjacent slices exhibit remarkable similarity. Various TV regularization [58], [59], [60], [61], [62], [63] uses the l_1-norm to constrain the difference images of adjacent slices. Although this TV regularization can explore the sparsity of difference images from a spectral or spatial perspective to some extent, they overlook the STCP. As shown in Fig. 1(c), each column of the horizontal and spatial difference images is either all the zeros

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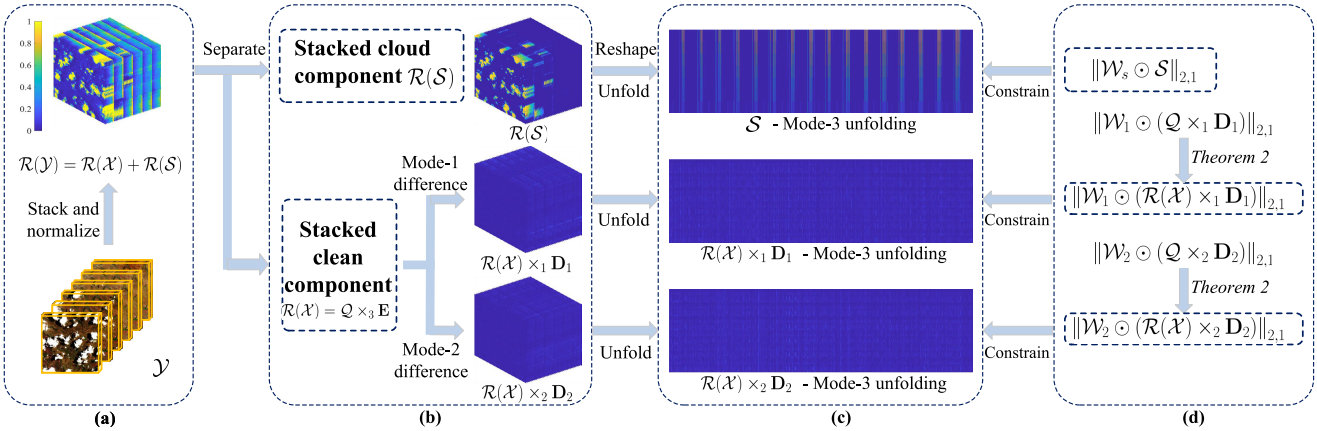


Fig. 1. The proposed STCP characterized by group sparsity regularization on spatial factor $\mathcal{Q}$ and cloud component $\mathcal{S}$. (a) Observed *Australia* data $\mathcal{Y} \in \mathbb{R}^{500\times500\times7\times6}$ and stacked data $\mathcal{R}(\mathcal{Y}) \in \mathbb{R}^{500\times500\times42}$. (b) Separated stacked cloud component $\mathcal{R}(\mathcal{S})$ and clean component $\mathcal{R}(\mathcal{X})$ which can be factorized as $\mathcal{Q} \times_3 \mathbf{E}$, where $\mathcal{Q} \in \mathbb{R}^{500\times500\times r}$, $\mathbf{E} \in \mathbb{R}^{42\times r}$, and $r \ll 42$. (c) Mode-3 unfolding results of the cloud component and difference images. (d) Regularization on cloud component $\mathcal{S}$ and regularization on spatial factor $\mathcal{Q}$ which replaces regularization on clean component $\mathcal{R}(\mathcal{X})$.

or close to zero, and smooth regions occur in similar positions. We introduce weighted group sparsity regularization to characterize the STCP, and the $l_{2,1}$-norm-based regularization function is represented as

$$\mathcal{G}(\mathcal{X}) = \sum_{k=1}^{2} \|\mathcal{W}_k \odot (\mathcal{R}(\mathcal{X}) \times_k \mathbf{D}_k)\|_{2,1}, \quad k = 1, 2 \tag{2}$$

where $\mathcal{W}_k$ is the weight tensor which can promote the group sparsity of spatial difference images of $\mathcal{R}(\mathcal{X})$, and $\mathbf{D}_k \in \mathbb{R}^{n_k \times n_k}$ is the difference matrix along the kth dimension.

However, due to the large size of MTRSIs, directly regularizing $\mathcal{R}(\mathcal{X}) \times_k \mathbf{D}_k$ in (2) would result in extremely high computational cost. To address this problem, we introduce a group sparsity regularization on spatial factor within the LRTF framework.

Factor-Based Group Sparsity Regularization: The stacked clean component $\mathcal{R}(\mathcal{X})$ can be approximated as follows:

$$\mathcal{R}(\mathcal{X}) = \mathcal{Q} \times_3 \mathbf{E} \tag{3}$$

where $\mathcal{Q} \in \mathbb{R}^{n_1\times n_2\times r}$, $\mathbf{E} \in \mathbb{R}^{n_3 n_4 \times r}$, and $r \ll \min(n_1, n_2, n_3 n_4)$. As the size of the spatial factor $\mathcal{Q}$ is much smaller than the stacked clean component $\mathcal{R}(\mathcal{X})$, we expect that $\mathcal{Q}$-based group sparsity regularization have a comparable effect to that of $\mathcal{R}(\mathcal{X})$-based regularization. Theorem 2 illustrates that applying group sparsity constraint on $\mathcal{Q}$ has the same effect as applying group sparsity constraint on $\mathcal{R}(\mathcal{X})$.

Theorem 2 (Equivalent Group Sparsity Constraint for Spatial Factor): Let $\mathcal{R}(\mathcal{X}) = \mathcal{Q} \times_3 \mathbf{E}$ be a third-order tensor and $\mathbf{D}_k$ be a matrix. Then $\mathcal{R}(\mathcal{X}) \times_k \mathbf{D}_k$ exhibits group sparsity along the third mode when $\mathcal{Q} \times_k \mathbf{D}_k$ exhibits group sparsity along the third mode.

Proof: The mode-k product of $\mathcal{R}(\mathcal{X})$ and $\mathbf{D}_k$ has the following law:

$$\mathcal{R}(\mathcal{X}) \times_k \mathbf{D}_k = (\mathcal{Q} \times_3 \mathbf{E}) \times_k \mathbf{D}_k = (\mathcal{Q} \times_k \mathbf{D}_k) \times_3 \mathbf{E}$$
$$\Leftrightarrow (\mathcal{R}(\mathcal{X}) \times_k \mathbf{D}_k)_{(i,j,:)} = \sum_{l=1}^{r} (\mathcal{Q} \times_k \mathbf{D}_k)_{(i,j,l)} \mathbf{E}_{(:,l)}$$

where r is the rank set for the low-rank approximation representation. The above derivation indicates that

$$(\mathcal{Q} \times_k \mathbf{D}_k)_{(i,j,:)} = 0 \Rightarrow (\mathcal{R}(\mathcal{X}) \times_k \mathbf{D}_k)_{(i,j,:)} = 0.$$

Based on Theorem 2, we replace the group sparsity regularization (2) on large-sized MTRSIs by the following factor-based regularization:

$$\mathcal{G}(\mathcal{Q}) = \sum_{k=1}^{2} \|\mathcal{W}_k \odot (\mathcal{Q} \times_k \mathbf{D}_k)\|_{2,1}, \quad k = 1, 2. \tag{4}$$

The regularization (4) on the small-sized factor can reduce computational complexity while capturing the STCP. Furthermore, considering that cloud components exhibit the same group sparsity, we use the same group sparsity constraint on the cloud component. The group sparsity regularization function on cloud component $\mathcal{S}$ is represented as

$$\mathcal{G}_s(\mathcal{S}) = \|\mathcal{W}_s \odot \mathcal{S}\|_{2,1} \tag{5}$$

where $\mathcal{W}_s$ is the weight tensor used to better promote the group sparsity of cloud component $\mathcal{S}$. By combining group sparsity functions (4) and (5), the MTRSIs cloud removal model using STCP can be formulated as follows:

$$\begin{aligned} \min_{\mathbf{E},\mathcal{Q},\mathcal{S},\mathcal{X}} & \sum_{k=1}^{2} \|\mathcal{W}_k \odot (\mathcal{Q} \times_k \mathbf{D}_k)\|_{2,1} \\ & + \frac{\lambda}{2} \|\mathcal{X} \times_4 \mathbf{D}_4\|_F^2 + \mu \|\mathcal{W}_s \odot \mathcal{S}\|_{2,1} \\ \text{s.t. } & \mathcal{R}(\mathcal{X}) = \mathcal{Q} \times_3 \mathbf{E}, \quad \mathcal{R}(\mathcal{Y}) = \mathcal{Q} \times_3 \mathbf{E} + \mathcal{R}(\mathcal{S}). \end{aligned} \tag{6}$$

where $\mu > 0$ and $\lambda > 0$ are regularization parameters, respectively.

B. Optimization Algorithm

We introduce two auxiliary variables $\mathcal{P}_1$ and $\mathcal{P}_2$ and rewrite problem (6) as

$$\begin{aligned} \min_{\mathbf{E},\mathcal{Q},\mathcal{P}_k,\mathcal{S},\mathcal{X}} & \sum_{k=1}^{2} \|\mathcal{W}_k \odot \mathcal{P}_k\|_{2,1} \\ & + \frac{\lambda}{2} \|\mathcal{X} \times_4 \mathbf{D}_4\|_F^2 + \mu \|\mathcal{W}_s \odot \mathcal{S}\|_{2,1} \\ \text{s.t. } & \mathcal{R}(\mathcal{X}) = \mathcal{Q} \times_3 \mathbf{E}, \quad \mathcal{R}(\mathcal{Y}) = \mathcal{Q} \times_3 \mathbf{E} + \mathcal{R}(\mathcal{S}) \\ & \mathcal{Q} \times_k \mathbf{D}_k = \mathcal{P}_k, \quad k = 1, 2. \end{aligned} \tag{7}$$

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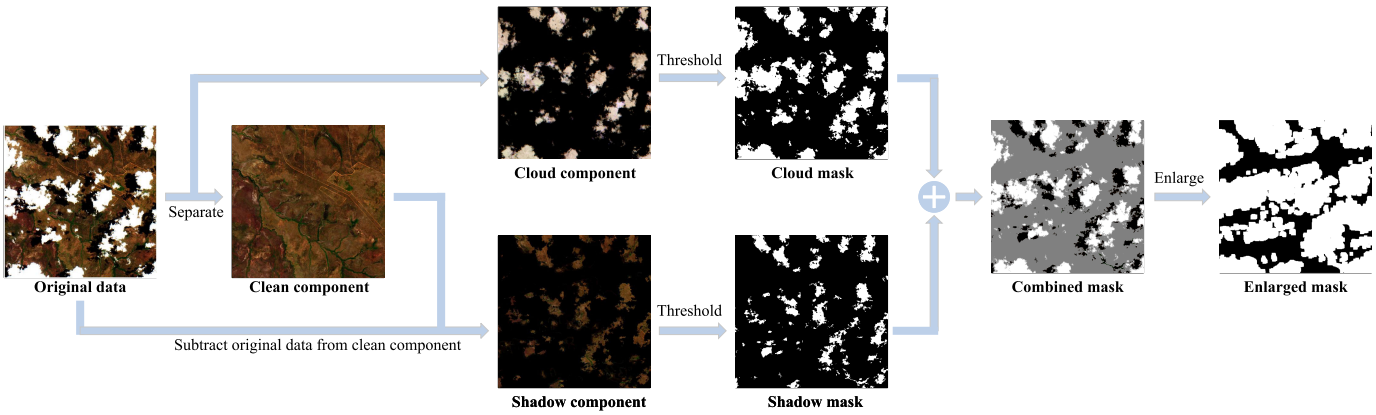


Fig. 2. Flowchart of the proposed cloud/shadow mask detection method, where $\oplus$ denotes combining cloud and shadow masks.

Based on the penalty method, the above optimization problem (7) can be converted into the following unconstrained function:

$$\begin{aligned} &\boldsymbol{f}(\mathbf{E}, \mathcal{Q}, \mathcal{P}_k, \mathcal{S}, \mathcal{X}) \\ &= \sum_{k=1}^{2} \|\mathcal{W}_k \odot \mathcal{P}_k\|_{2,1} + \frac{\alpha}{2} \sum_{k=1}^{2} \|\mathcal{P}_k - \mathcal{Q} \times_k \mathbf{D}_k\|_F^2 \\ &\quad + \mu \|\mathcal{W}_s \odot \mathcal{S}\|_{2,1} + \frac{\sigma_1}{2} \|\mathcal{R}(\mathcal{Y}) - \mathcal{Q} \times_3 \mathbf{E} - \mathcal{R}(\mathcal{S})\|_F^2 \\ &\quad + \frac{\lambda}{2} \|\mathcal{X} \times_4 \mathbf{D}_4\|_F^2 + \frac{\sigma_2}{2} \|\mathcal{R}(\mathcal{X}) - \mathcal{Q} \times_3 \mathbf{E}\|_F^2 \end{aligned} \tag{8}$$

where $\alpha > 0$ and $\sigma_i > 0$ $(i = 1, 2)$ are penalty parameters.

We use the PAM-based algorithm to solve problem (8), whose solution is obtained by alternately updating

$$\begin{cases} \mathbf{E} = \underset{\mathbf{E}}{\operatorname{argmin}}\, \boldsymbol{f}(\mathbf{E}, \mathcal{Q}, \mathcal{P}_k, \mathcal{S}, \mathcal{X}) + \frac{\beta}{2} \|\mathbf{E} - \hat{\mathbf{E}}\|_F^2 \\ \mathcal{Q} = \underset{\mathcal{Q}}{\operatorname{argmin}}\, \boldsymbol{f}(\mathbf{E}, \mathcal{Q}, \mathcal{P}_k, \mathcal{S}, \mathcal{X}) + \frac{\beta}{2} \|\mathcal{Q} - \hat{\mathcal{Q}}\|_F^2 \\ \mathcal{P}_k = \underset{\mathcal{P}_k}{\operatorname{argmin}}\, \boldsymbol{f}(\mathbf{E}, \mathcal{Q}, \mathcal{P}_k, \mathcal{S}, \mathcal{X}) + \frac{\beta}{2} \|\mathcal{P}_k - \hat{\mathcal{P}}_k\|_F^2 \\ \mathcal{S} = \underset{\mathcal{S}}{\operatorname{argmin}}\, \boldsymbol{f}(\mathbf{E}, \mathcal{Q}, \mathcal{P}_k, \mathcal{S}, \mathcal{X}) + \frac{\beta}{2} \|\mathcal{S} - \hat{\mathcal{S}}\|_F^2 \\ \mathcal{X} = \underset{\mathcal{X}}{\operatorname{argmin}}\, \boldsymbol{f}(\mathbf{E}, \mathcal{Q}, \mathcal{P}_k, \mathcal{S}, \mathcal{X}) + \frac{\beta}{2} \|\mathcal{X} - \hat{\mathcal{X}}\|_F^2 \end{cases} \tag{9}$$

where $\hat{\mathbf{E}}$, $\hat{\mathcal{Q}}$, $\hat{\mathcal{P}}_k$, $\hat{\mathcal{S}}$, and $\hat{\mathcal{X}}$ are the solutions of the corresponding subproblems at the previous iteration, respectively, and $\beta > 0$ is the proximal parameter.

1) Updating $\boldsymbol{E}$: $\mathbf{E}$-subproblem is as follows:

$$\begin{aligned} \min_{\mathbf{E}}\ & \frac{\sigma_1}{2} \|\mathcal{R}(\mathcal{Y}) - \mathcal{Q} \times_3 \mathbf{E} - \mathcal{R}(\mathcal{S})\|_F^2 \\ & + \frac{\sigma_2}{2} \|\mathcal{R}(\mathcal{X}) - \mathcal{Q} \times_3 \mathbf{E}\|_F^2 + \frac{\beta}{2} \left\|\mathbf{E} - \hat{\mathbf{E}}\right\|_F^2. \end{aligned} \tag{10}$$

The objective function of (10) is differentiable, and thus its solution can be obtained by

$$\begin{aligned} \mathbf{E} = & \left((\sigma_1 \mathcal{R}(\mathcal{Y}) + \sigma_2 \mathcal{R}(\mathcal{X}) - \sigma_1 \mathcal{R}(\mathcal{S}))_{(3)} \mathbf{Q}_{(3)}^{\mathsf{T}} + \beta \hat{\mathbf{E}}\right) \\ & \times \left((\sigma_1 + \sigma_2) \mathbf{Q}_{(3)} \mathbf{Q}_{(3)}^{\mathsf{T}} + \beta \mathbf{I}\right)^{-1}. \end{aligned} \tag{11}$$

2) Updating $\mathcal{Q}$: $\mathcal{Q}$-subproblem is as follows:

$$\begin{aligned} \min_{\mathcal{Q}}\ & \frac{\alpha}{2} \sum_{k=1}^{2} \|\mathcal{P}_k - \mathcal{Q} \times_k \mathbf{D}_k\|_F^2 + \frac{\beta}{2} \left\|\mathcal{Q} - \hat{\mathcal{Q}}\right\|_F^2 \\ & + \frac{\sigma_1}{2} \|\mathcal{R}(\mathcal{Y}) - \mathcal{Q} \times_3 \mathbf{E} - \mathcal{R}(\mathcal{S})\|_F^2 \\ & + \frac{\sigma_2}{2} \|\mathcal{R}(\mathcal{X}) - \mathcal{Q} \times_3 \mathbf{E}\|_F^2. \end{aligned} \tag{12}$$

The solution of problem (12) can be obtained by

$$\begin{aligned} & (\sigma_1 + \sigma_2) \mathcal{Q} \times_3 \left(\mathbf{E}^{\mathsf{T}} \mathbf{E}\right) \\ & \qquad + \alpha \sum_{k=1}^{2} \mathcal{Q} \times_k (\mathbf{D}_k^{\mathsf{T}} \mathbf{D}_k) + \beta \mathcal{Q} = \mathcal{A} \end{aligned} \tag{13}$$

where

$$\begin{aligned} \mathcal{A} = & (\sigma_1 \mathcal{R}(\mathcal{Y}) + \sigma_2 \mathcal{R}(\mathcal{X}) - \sigma_1 \mathcal{R}(\mathcal{S})) \times_3 \mathbf{E}^{\mathsf{T}} \\ & + \alpha \sum_{k=1}^{2} \mathcal{P}_k \times_k \mathbf{D}_k^{\mathsf{T}} + \beta \hat{\mathcal{Q}}. \end{aligned}$$

To use the fast solution of the Sylvester matrix equation mentioned in Theorem 1, we write problem (13) in accordance to Properties 2–6 in [64] as the following mode-3 unfolding form:

$$(\sigma_1 + \sigma_2) \mathbf{Q}_{(3)}^{\mathsf{T}} \left(\mathbf{E}^{\mathsf{T}} \mathbf{E}\right) + \beta \mathbf{Q}_{(3)}^{\mathsf{T}} + \mathbf{B} \mathbf{Q}_{(3)}^{\mathsf{T}} = \mathbf{A}_{(3)}^{\mathsf{T}} \tag{14}$$

where $\mathbf{B} = \alpha[(\mathbf{I}_{n_2} \otimes \mathbf{D}_1^{\mathsf{T}} \mathbf{D}_1) + (\mathbf{D}_2^{\mathsf{T}} \mathbf{D}_2 \otimes \mathbf{I}_{n_1})]$. We use fast Fourier transform and singular value decomposition to diagonalize $\mathbf{B}$ and $\mathbf{E}^{\mathsf{T}}\mathbf{E}$, respectively. That is,

$$\mathbf{B} = \mathbf{F}^{\mathsf{T}} \Psi \mathbf{F} \quad \text{and} \quad \mathbf{E}^{\mathsf{T}} \mathbf{E} = \mathbf{U} \Sigma \mathbf{U}^{\mathsf{T}} \tag{15}$$

where $\mathbf{F}$ is the 2-D discrete Fourier transformation matrix, and Ψ and Σ are the diagonal matrices. As (14) conforms to the Sylvester matrix equation structure, we can obtain the fast solution of $\mathcal{Q}$

$$\mathcal{Q} = \operatorname{Fold}_3\left(\left[\mathbf{F}^{\mathsf{T}}\left((1 \oslash \mathbf{T}) \odot (\mathbf{F} \mathbf{A}_{(3)}^{\mathsf{T}} \mathbf{U})\right) \mathbf{U}^{\mathsf{T}}\right]^{\mathsf{T}}\right) \tag{16}$$

where

$$\begin{aligned} \mathbf{T} = & \beta \operatorname{ones}(n_1 n_2, r) + (\operatorname{diag}(\Psi), \operatorname{diag}(\Psi), \ldots, \operatorname{diag}(\Psi)) \\ & + (\sigma_1 + \sigma_2)(\operatorname{diag}(\Sigma), \operatorname{diag}(\Sigma), \ldots, \operatorname{diag}(\Sigma))^{\mathsf{T}} \end{aligned}$$

and $\operatorname{ones}(n_1 n_2, r)$ is a matrix of size $n_1 n_2 \times r$ whose elements are all ones.

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3) Updating $\mathcal{P}_k$: $\mathcal{P}_k$-subproblem is as follows:

$$\min_{\mathcal{P}_k} \sum_{k=1}^{2} \|\mathcal{W}_k \odot \mathcal{P}_k\|_{2,1} + \frac{\alpha}{2}\sum_{k=1}^{2}\|\mathcal{P}_k - \mathcal{Q} \times_k \mathbf{D}_k\|_F^2 + \frac{\beta}{2}\sum_{k=1}^{2}\left\|\mathcal{P}_k - \hat{\mathcal{P}}_k\right\|_F^2. \tag{17}$$

The optimization problem (17) can be solved by the soft shrinkage operator which is defined as

$$\text{shrink}_{2,1}(\mathbf{x}, \xi) = \begin{cases} \dfrac{\|\mathbf{x}\|_2 - \xi}{\|\mathbf{x}\|_2}\mathbf{x}, & \text{if } \xi < \|\mathbf{x}\|_2 \\ 0, & \text{otherwise.} \end{cases} \tag{18}$$

All the tubes of $\mathcal{P}_k$ have a consistent direct solution of the following formula:

$$(\mathcal{P}_k)_{(i,j,:)} = \text{shrink}_{2,1}\left((\mathcal{M}_k)_{(i,j,:)}, \frac{|(\mathbf{W}_k)_{(i,j)}|}{\alpha + \beta}\right) \tag{19}$$

where $\mathcal{M}_k = (1/(\alpha + \beta))(\alpha\mathcal{Q} \times_k \mathbf{D}_k + \beta\hat{\mathcal{P}}_k)$ units all the variables except $\mathcal{P}_k$ in problem (17); $\mathbf{W}_k$ is the frontal slice of tensor $\mathcal{W}_k$, which satisfies $(\mathbf{W}_k)_{(i,j)} = (1/(\|(\mathcal{M}_k)_{(i,j,:)}\|_2 + \varphi))$ and φ is a minuscule constant.

4) Updating $\mathcal{S}$: $\mathcal{S}$-subproblem is as follows:

$$\min_{\mathcal{S}} \mu\|\mathcal{W}_s \odot \mathcal{S}\|_{2,1} + \frac{\beta}{2}\left\|\mathcal{S} - \hat{\mathcal{S}}\right\|_F^2 + \frac{\sigma_1}{2}\|\mathcal{R}(\mathcal{Y}) - \mathcal{Q} \times_3 \mathbf{E} - \mathcal{R}(\mathcal{S})\|_F^2. \tag{20}$$

All the tubes of $\mathcal{S}$ have a consistent direct solution of the following formula:

$$\mathcal{S}_{(i,j,:,l)} = \text{shrink}_{2,1}\left(\mathcal{N}_{(i,j,:,l)}, \frac{\mu|(\tilde{\mathcal{W}}_s)_{(i,j,l)}|}{\sigma_1 + \beta}\right) \tag{21}$$

where $\mathcal{N} = (\beta\hat{\mathcal{S}} + \sigma_1(\mathcal{Y} - \mathcal{R}^{-1}(\mathcal{Q} \times_3 \mathbf{E})))/(\beta + \sigma_1)$ replaces the variables except $\mathcal{S}$ in the fidelity and penalty terms in problem (20), $\mathcal{R}^{-1}(\cdot)$ is the inverse operator of $\mathcal{R}(\cdot)$, and the weight tensor $(\tilde{\mathcal{W}}_s)_{(i,j,l)} = (1/(|\mathcal{N}_{i,j,:,l}| + \varphi))$ is the subtensor of $\mathcal{W}_s$.

5) Updating $\mathcal{X}$: $\mathcal{X}$-subproblem is as follows:

$$\min_{\mathcal{X}} \frac{\sigma_2}{2}\|\mathcal{R}(\mathcal{X}) - \mathcal{Q} \times_3 \mathbf{E}\|_F^2 + \frac{\beta}{2}\left\|\mathcal{X} - \hat{\mathcal{X}}\right\|_F^2 + \frac{\lambda}{2}\|\mathcal{X} \times_4 \mathbf{D}_4\|_F^2. \tag{22}$$

The solution of $\mathcal{X}$ can be directly obtained by

$$\mathcal{X} = \text{Fold}_4\left(\left(\sigma_2\left(\mathcal{R}^{-1}(\mathcal{Q} \times_3 \mathbf{E})\right)_{(4)} + \beta\hat{\mathbf{X}}_{(4)}\right) \times \left(\lambda\mathbf{D}_4^{\top}\mathbf{D}_4 + (\beta + \sigma_2)\mathbf{I}\right)^{-1}\right). \tag{23}$$

Mask Detection and Result Refinement: To avoid inconsistency between the reconstructed and original images in cloud-free regions, we use a mask to optimize the target clean images $\mathcal{X}$ obtained from (23) at the end of the iterative algorithm. Fig. 2 displays the flowchart of the proposed cloud/shadow mask detection method. We summarize the aforementioned process as Algorithm 1.

Algorithm 1 PAM-Based Solving Algorithm for MTRSIs Cloud Removal Model

Input: The observed cloud-contained MTRSIs $\mathcal{Y}$ and parameters α, λ, μ, σ_1, σ_2;

1: **Initialize:** $\mathbf{E}^0 = \text{rand}(n_3n_4, r)$, $\mathcal{Q}^0 = \text{rand}(n_1, n_2, r)$, $\mathcal{P}_k^0 = \text{rand}(n_1, n_2, r)$, $\mathcal{S}^0 = 0$, $\mathcal{X}^0 = \mathcal{Y}$.
2: **while** not converged and iterations < 150 **do**
3: Update $\mathbf{E}$ via (11).
4: Update $\mathcal{Q}$ via (16).
5: Update $\mathcal{P}_k$ via (19), $k = 1, 2$.
6: Update $\mathcal{S}$ via (21).
7: Update $\mathcal{X}$ via (23).
8: Check the convergence condition: $\|\mathcal{X} - \hat{\mathcal{X}}\|_F / \|\hat{\mathcal{X}}\|_F < 10^{-4}$.
9: **end while**
10: Calculate Ω_c via thresholding on $\mathcal{S}$.
11: Calculate Ω_s via thresholding on $\mathcal{X} - \mathcal{Y}$.
12: Refine $\mathcal{X}$ with masks Ω_c and Ω_s.

Output: reconstructed MTRSIs $\mathcal{X}$.

C. Computational Complexity Analysis

At each iteration in Algorithm 1, the main computational complexity lies in updating $\mathbf{E}$, $\mathcal{Q}$, $\mathcal{P}_k$, $\mathcal{S}$, and $\mathcal{X}$, where $k = 1, 2$. For the factors obtained by LRTF, the complexity of computing factor $\mathbf{E}$ is $\mathcal{O}(n_1n_2n_3n_4r + (n_1n_2 + n_3n_4)r^2)$ and the updating factor $\mathcal{Q}$ needs $\mathcal{O}((n_1 + n_2 + r)n_1n_2n_3n_4 + r^2n_3n_4 + rn_1n_2\log(n_1n_2))$. The computational complexity of updating variable $\mathcal{P}_k$ is $\mathcal{O}(n_1n_2n_3n_4)$. For the separated cloud and clean components, the cost of computing $\mathcal{S}$ is $\mathcal{O}(n_1n_2n_3n_4)$, and the cost of computing $\mathcal{X}$ is $\mathcal{O}(n_1n_2n_3n_4(r + n_4))$. In conclusion, the complexity at each iteration is $\mathcal{O}((r + n_1 + n_2 + n_4)n_1n_2n_3n_4 + (n_1n_2 + n_3n_4)r^2 + rn_1n_2\log(n_1n_2))$.

IV. Experiments

In this section, we test the performance of the proposed STCP method on both synthetic and real experiments. We select six methods for comparison including TVLRSDC [65], RTCR [51], SLRpGS [66], MT [67], BSN [68], and TRGFid [38]. The brief overview of these comparative methods is displayed in Table I. We adjust all the parameters of the aforementioned methods to achieve optimal performance according to the authors' code and suggestions. We apply the same mask acquired by the proposed method to refine cloud removal results of all the mask-unknown methods in the comparison. The experiments are implemented under Windows 11 and MATLAB (R2022b) running on a desktop PC with an Intel Core i7-13700F CPU at 2.10 GHz and 32.0-GB RAM. We use four figures of merit, peak signal-to-noise rate (PSNR), structural similarity index measure (SSIM), spectral angle mapping (SAM), and correlation coefficient (CC), to evaluate the quality of the reconstructed results. The PSNR, SSIM, SAM, and CC are defined as follows:

$$\text{PSNR} = \frac{1}{n_3n_4}\sum_{b=1}^{n_3}\sum_{t=1}^{n_4} 10\log\frac{n_1n_2}{\left\|\mathbf{X}_b^t - \hat{\mathbf{X}}_b^t\right\|_F^2}$$

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TABLE I
BRIEF INFORMATION OF ALL COMPARED AND PROPOSED METHODS

Method	Category	Low-rank prior	Local continuity prior	spectral-temporal consistency prior
TVLRSDC [65]	Mask-unknown/tensor model	Matrix rank	Spatial and spectral smooth	-
RTCR [51]	Semi-mask/tensor model	Matrix rank	-	-
SLRpGS [66]	Mask-unknown/tensor model	Multi rank	Temporal smooth	Regular on cloud component
MT [67]	Mask-known/deep learning	-	-	-
BSN [68]	Mask-known/deep learning	-	-	-
TRGFid [38]	Mask-known/tensor model	TR rank	Gradient domain smooth	-
proposed STCP	Mask-unknown/tensor model	Tucker/matrix rank	Temporal smooth	Weighted regular on spatial factor and cloud component

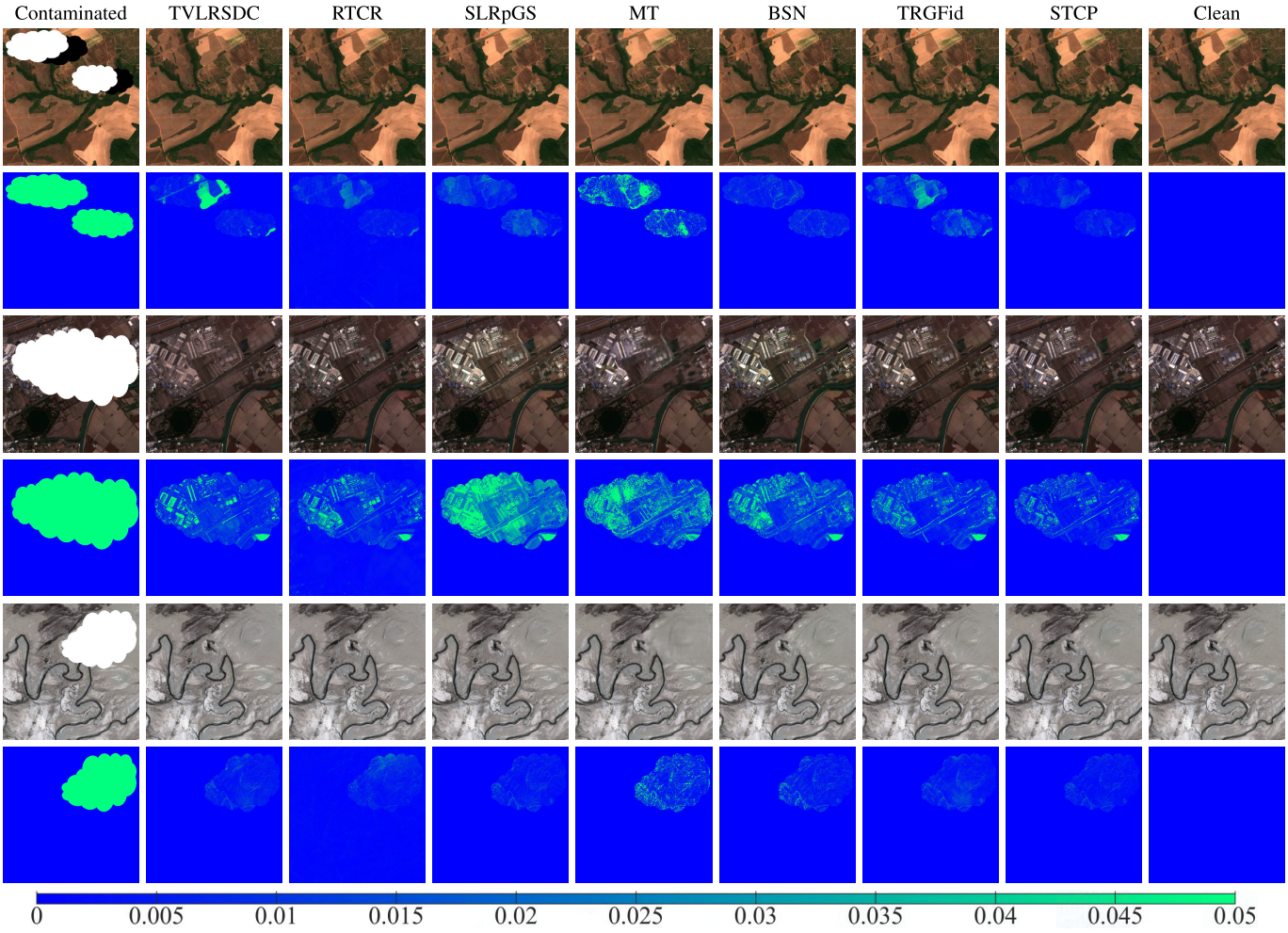


Fig. 3. Reconstruction results for time node 1 of the *Brazil* dataset, node 2 of the *Italy* dataset, and node 3 of the *China* dataset by all compared and proposed methods. Rows 1, 3, and 5: visual renderings (B4, B3, and B2 represent the R, G, and B values). Rows 2, 4, and 6: the corresponding residual maps.

$$\mathrm{SSIM} = \frac{1}{n_3 n_4}\sum_{b=1}^{n_3}\sum_{t=1}^{n_4}\frac{\left(2\mu_{\mathbf{X}_b^t}\mu_{\hat{\mathbf{X}}_b^t}+\varphi_1\right)\left(2\sigma_{\mathbf{X}_b^t\hat{\mathbf{X}}_b^t}+\varphi_2\right)}{\left(\mu_{\mathbf{X}_b^t}^2+\mu_{\hat{\mathbf{X}}_b^t}^2+\varphi_1\right)\left(\sigma_{\mathbf{X}_b^t}^2+\sigma_{\hat{\mathbf{X}}_b^t}^2+\varphi_2\right)}$$

$$\mathrm{SAM} = \frac{1}{n_1 n_2 n_4}\sum_{n_1,n_2,n_4}\cos^{-1}\left(\frac{\left(\mathbf{X}_{i,j}^t\right)^{\mathrm{T}}\hat{\mathbf{X}}_{i,j}^t}{\left\|\mathbf{X}_{i,j}^t\right\|_F\left\|\hat{\mathbf{X}}_{i,j}^t\right\|_F+\varphi_3}\right)$$

$$\mathrm{CC} = \frac{\sum_{k=1}^{K}\left(\mathcal{X}_k-\mu_{\mathcal{X}}\right)\left(\hat{\mathcal{X}}_k-\mu_{\hat{\mathcal{X}}}\right)}{\sqrt{\sum_{k=1}^{K}\left(\mathcal{X}_k-\mu_{\mathcal{X}}\right)^2\sum_{k=1}^{K}\left(\hat{\mathcal{X}}_k-\mu_{\hat{\mathcal{X}}}\right)^2}}$$

where n_1, n_2, n_3, n_4 denote the horizontal and vertical lengths, number of bands, and time nodes, respectively; $\mathbf{X}$ denotes the reconstructed image; $\hat{\mathbf{X}}$ denotes the ground truth; subscripts b and t denote the bth band and tth time node, respectively; $\mu_{\mathbf{X}_b^t}$ and $\mu_{\hat{\mathbf{X}}_b^t}$ denote the mean values of $\mathbf{X}_b^t$ and $\hat{\mathbf{X}}_b^t$, respectively; $\sigma_{\mathbf{X}_b^t}^2$ and $\sigma_{\hat{\mathbf{X}}_b^t}^2$ denote the standard deviation of $\mathbf{X}_b^t$ and $\hat{\mathbf{X}}_b^t$, respectively; $\sigma_{\mathbf{X}_b^t\hat{\mathbf{X}}_b^t}$ denotes the covariance of $\mathbf{X}_b^t$ and $\hat{\mathbf{X}}_b^t$; $\mathbf{X}_{i,j}^t$ and $\hat{\mathbf{X}}_{i,j}^t$ are tubes by fixing indexes i, j, t; $\mathcal{X}_k$ and $\hat{\mathcal{X}}_k$ are the kth pixel values of reconstructed images and ground truth, respectively; $\mu_{\mathcal{X}}$ and $\mu_{\hat{\mathcal{X}}}$ are the corresponding mean values, respectively; K denotes the number of pixels in the reconstructed images; and φ_1, φ_2, and φ_3 are constants. The higher values of PSNR, SSIM, and CC indicate better reconstruction quality while that of SAM indicates poorer. All test data are available at https://dataspace.copernicus.eu/.

A. Synthetic Experiments

We use three MTRSIs' datasets to test the performance of the proposed STCP method in synthetic experiments.

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TABLE II

SETTING OF SIMULATED CLOUD AND SHADOW IN DIFFERENT TEMPORAL NODES OF THREE DATASETS

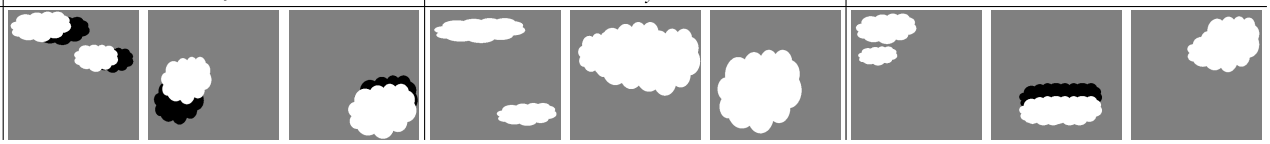
Datasets	*Brazil*			*Italy*			*China*		
Mask									

TABLE III

QUANTITATIVE RESULTS OF COMPARED AND PROPOSED METHODS ON SYNTHETIC EXPERIMENTS

Datasets	Time node	Index	Observed	TVLRSDC	RTCR	SLRpGS	MT	BSN	TRGFid	STCP
Brazil	Time node 1	PSNR	10.5869	35.1400	44.9535	44.6128	38.8497	43.7084	41.9061	**50.0332**
		SSIM	0.8101	0.9882	0.9954	0.9963	0.9730	0.9855	0.9945	**0.9973**
		SAM	9.2298	0.3290	0.4077	0.2176	0.4288	0.5013	0.2338	**0.1580**
		CC	0.4035	0.9925	0.9979	0.9992	0.9915	0.9827	0.9985	**0.9998**
	Time node 2	PSNR	11.7034	45.2306	48.6259	42.2215	45.3222	42.2305	47.4275	**49.5535**
		SSIM	0.8438	0.9950	0.9966	0.9956	0.9890	0.9891	0.9955	**0.9971**
		SAM	8.6882	0.2254	0.3923	0.2560	0.3016	0.2157	0.2197	**0.1782**
		CC	0.4534	0.9991	0.9990	0.9987	0.9983	0.9932	0.9996	**0.9998**
	Time node 3	PSNR	10.8369	38.4585	43.9881	46.9703	40.1458	41.3646	43.5012	**47.9834**
		SSIM	0.8390	0.9873	0.9940	0.9949	0.9805	0.9771	0.9915	**0.9955**
		SAM	7.7015	0.3244	0.4551	0.1860	0.3568	0.7570	0.2191	**0.1576**
		CC	0.5354	0.9967	0.9978	0.9996	0.9946	0.9716	0.9991	**0.9997**
		Time (s)	-	73.1857	128.7634	192.3767	344.5647	3187.2441	47.9302	**21.3025**
Italy	Time node 1	PSNR	9.3660	39.4836	40.4071	33.6086	38.9763	36.8509	39.9379	**41.4076**
		SSIM	0.8291	0.9873	0.9878	0.9774	0.9687	0.9797	0.9848	**0.9893**
		SAM	3.0228	0.5845	1.3355	0.5667	0.4830	1.0538	0.5470	**0.4459**
		CC	0.3181	0.9920	0.9902	0.9754	0.9849	0.9795	0.9925	**0.9947**
	Time node 2	PSNR	5.7616	33.8183	36.7910	32.2478	32.8380	34.0170	36.8300	**39.1022**
		SSIM	0.6347	0.9679	0.9804	0.9686	0.9111	0.9679	0.9694	**0.9806**
		SAM	8.2251	1.6365	1.4712	1.4609	1.5022	1.3343	1.2622	**0.8188**
		CC	0.2536	0.9744	0.9845	0.9750	0.9487	0.9690	0.9869	**0.9924**
	Time node 3	PSNR	6.5589	34.3670	38.2212	37.9920	27.2516	38.9861	38.3815	**40.4418**
		SSIM	0.7018	0.9704	0.9852	0.9829	0.8956	0.9820	0.9753	**0.9861**
		SAM	9.0056	2.4645	1.5861	1.2484	1.2979	0.9142	1.0859	**0.7178**
		CC	0.0459	0.9836	0.9895	0.9924	0.8614	0.9903	0.9931	**0.9956**
		Time (s)	-	131.0386	122.3103	188.1112	347.7089	3360.2603	136.4596	**23.0576**
China	Time node 1	PSNR	11.6946	**41.7812**	40.5882	40.2078	34.1340	39.6383	41.1717	41.0350
		SSIM	0.8927	0.9882	0.9891	0.9886	0.9761	0.9865	0.9888	**0.9892**
		SAM	0.8207	0.3120	0.5046	0.1739	0.1863	0.1918	0.2045	**0.1652**
		CC	0.1065	0.9904	0.9900	0.9897	0.9608	0.9870	**0.9917**	0.9914
	Time node 2	PSNR	11.8161	42.1679	43.1754	42.8550	40.1484	41.5615	46.5722	**47.8371**
		SSIM	0.8300	0.9945	0.9947	0.9957	0.9708	0.9917	0.9955	**0.9967**
		SAM	6.0259	0.2962	0.5714	0.1903	0.1896	0.2633	0.1879	**0.1651**
		CC	0.3527	0.9942	0.9955	0.9958	0.9915	0.9931	0.9979	**0.9984**
	Time node 3	PSNR	10.6212	48.7360	48.3025	47.0037	43.2316	48.5894	49.8450	**51.5615**
		SSIM	0.8766	0.9966	0.9962	0.9939	0.9782	0.9943	0.9968	**0.9969**
		SAM	0.5535	0.1999	0.4276	0.2003	0.1382	0.1609	0.1381	**0.1145**
		CC	0.2830	0.9984	0.9984	0.9980	0.9954	0.9985	0.9989	**0.9993**
		Time (s)	-	89.5750	120.6319	85.8425	420.6019	3243.5837	49.9073	**31.5893**
	Average	PSNR	9.8828	39.9092	42.7837	40.8577	37.8775	40.7719	42.8414	**45.4395**
		SSIM	0.8064	0.9861	0.9910	0.9882	0.9603	0.9838	0.9880	**0.9921**
		SAM	5.9192	0.7080	0.7946	0.5000	0.5424	0.5991	0.4553	**0.3246**
		CC	0.3057	0.9912	0.9936	0.9915	0.9697	0.9850	0.9953	**0.9968**
		Time (s)	-	97.9331	123.9019	155.4435	370.9585	3263.6960	78.0990	**25.3165**

1) Simulated Datasets: To demonstrate the capability of the proposed STCP method in cloud removal tasks for MTRSIs of various terrains, we select images data of fields, cities, and wastelands. The first dataset, *Brazil*, was captured in Gurupi, Brazil, in June, July, and August 2020. The second dataset, *Italy*, was captured in Rome, Italy, in August, September, and October 2023. The third dataset, *China*, was captured in Qinghai, China, in July, August, and September 2023. All the data were obtained from Sentinel-2 and cropped to a spatial size of 400 × 400. They all have four spectral bands (B2, B3, B4, and B8) and six time nodes. We add simulated clouds and shadows to the images of the first three time nodes. All the spectral bands of the same time node are contaminated by the same clouds and shadows, and the

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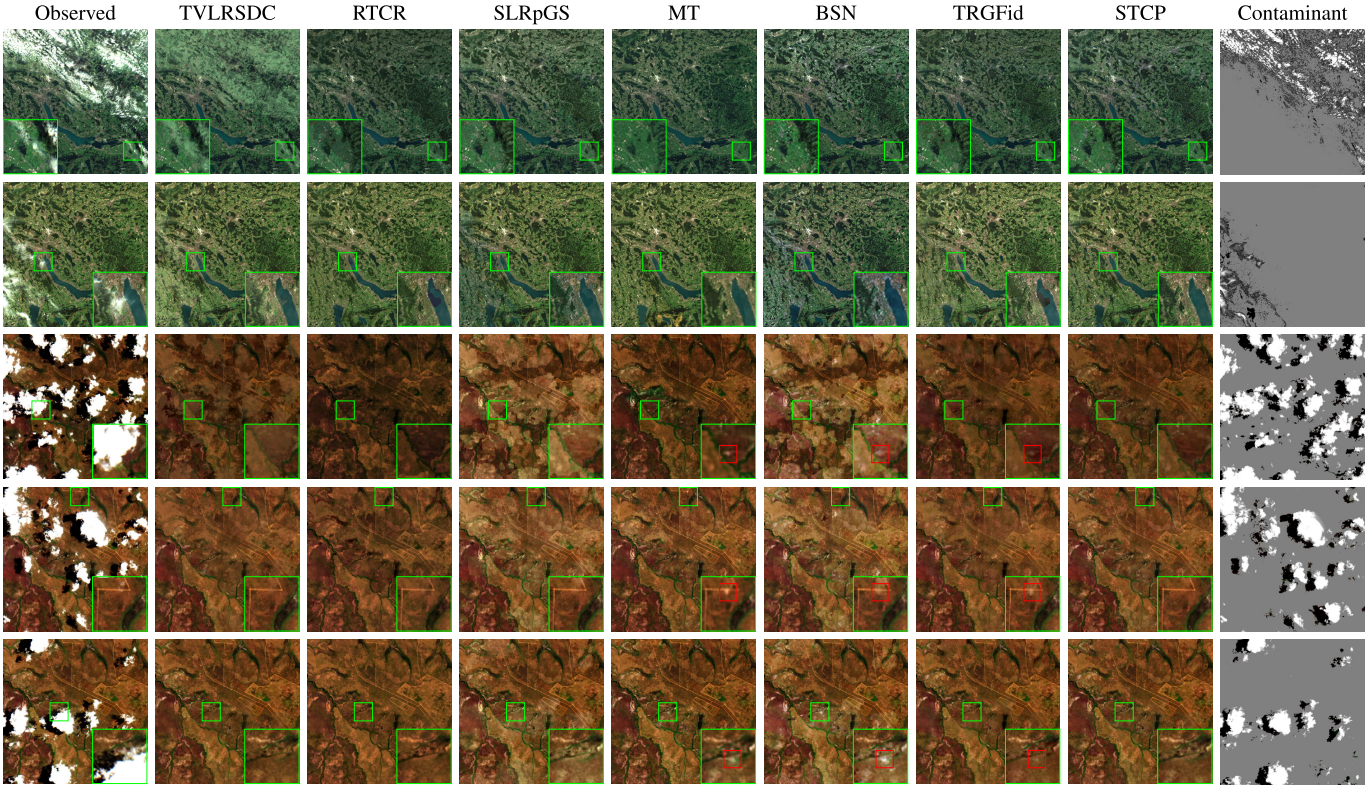


Fig. 4. Observed data, reconstruction results (B4, B3, and B2 represent the R, G, and B values), and detected cloud/shadow on the *Switzerland* and *Australia* datasets, respectively.

images at different time nodes are contaminated by clouds and shadows of different sizes, positions, and shapes. Table II shows the specific situations of adding simulated clouds and shadows to the mentioned MTRSIs.

2) Experimental Results: We present in Table III the performance metrics' evaluation of STCP and all the comparative methods in the synthetic experiments, where **bold** denotes best values and underlined denotes suboptimal values. The STCP method achieves the best overall results in terms of evaluation metrics in a majority of synthetic experiments and exhibits significant improvements compared with the suboptimal method. Furthermore, STCP demonstrates a considerably shorter average runtime compared with other mask-unknown and mask-known methods. Particularly, compared with the deep learning method BSN, which requires training, STCP has a speed advantage of over 100 times. The superior performance of STCP can be attributed to effective utilization of the STCP and its fast performance is due to applying the regularization on the spatial factor rather than the original images. To qualitatively compare the performance of STCP and all the comparative methods in the simulated data experiments, we present in rows 1, 3, and 5 of Fig. 3 the visual results of cloud removal for the first time node of *Brazil*, the second time node of *Italy*, and the third time node of *China*, respectively. In rows 2, 4, and 6 of Fig. 3, we show the corresponding residual maps to provide a more intuitive comparison. The larger the difference between the cloud removal result and the true cloud-free image, the closer the color is to green; conversely, the closer the color is to blue. It can be observed that the reconstruction results of TVLRSDC, RTCR, and SLRpGS exhibit varying degrees of color differences and artifacts. The reconstruction results of BSN, MT, and TRGFid still have shortcomings in terms of detail and texture restoration. The STCP reconstruction images have the closest effect to clean images.

B. Real Experiments

We use two MTRSIs' datasets to test the performance of the proposed STCP method in real experiments. The experimental results are presented in Fig. 4.

1) Real Datasets: Fig. 5 displays images of all the time nodes of the *Switzerland* and the *Australia* datasets, respectively. The first dataset, *Switzerland*, was captured in September and October 2023 in Zurich, Switzerland. The second dataset, *Australia*, was captured in September and October 2023 on Melville Island, Australia. Table IV presents detailed information about these two datasets. We use the first three time nodes of the *Australia* dataset and the first two time nodes of the *Switzerland* dataset as the target for cloud removal, with the remaining nodes serving as reference. It is worth noting that unlike the synthetic data, the reference images of real data are not completely free from cloud and shadow contamination.

2) Experimental Results: Fig. 4 presents the cloud removal results of STCP and all the comparative methods on the *Switzerland* and *Australia* datasets, as well as the cloud and shadow masks detected by STCP. To facilitate clear observation and comparison, we enlarge the regions indicated by the green boxes and mark the obvious clouds that are not removed with red boxes. For experiment on the *Switzerland*

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TABLE IV
DETAILED INFORMATION OF MTRSIS FOR REAL EXPERIMENTS

	Spatial resolution	Image size	Spectral bands	Temporal nodes
Switzerland	60m	1000×1000	4	4
Australia	20m	500×500	7	6

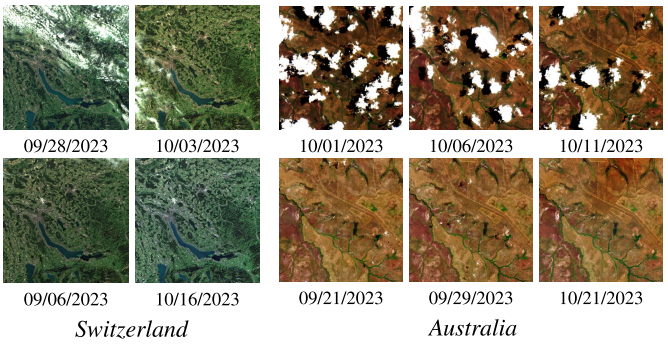


Fig. 5. (Left) *Switzerland* and (right) *Australia* color images (B4, B3, and B2 represent the R, G, and B values) of Sentinel-2 datasets.

dataset, the reconstruction results of TVLRSDC, MT, and BSN have obvious losses of texture and color details in large regions. RTCR, SLRpGS, and TRGFid fail to effectively overcome the edge color difference problem. STCP has a clear superiority in detail restoration and dealing with color difference issues. For experiment on the *Australia* dataset, although the reconstruction images of MT, BSN, and TRGFid have acceptable global effects, they are unable to remove the clouds from the reference images. The reconstruction images of TVLRSDC, RTCR, and SLRpGS have relatively severe color difference problems between the contaminated and uncontaminated regions. STCP can avoid the clouds in the reference image from appearing in the reconstruction regions while achieving the best reconstruction effect.

C. Discussions

We investigate the impact of parameters on cloud removal performance. The proposed STCP method involves six key parameters: rank r, penalty parameters α, σ_1, and σ_2, and regularization parameters λ and μ. Fig. 6 reflects the impact of parameters on the effectiveness of the model on the *Brazil* dataset.

1) Rank r: The variation in rank r has a significant effect on cloud removal performance of the model. From Fig. 6(a), it can be observed that the proposed model achieves stable performance when $r \geq 6$ on the *Brazil* dataset.

2) Penalty Parameters α, σ_1, and σ_2: The parameter α is used to balance the penalty term of $\mathcal{P}_k$ $(k = 1, 2)$. It can be observed that the proposed model yields the maximum PSNR and requires the fewest iterations when $\alpha = 10\,000$. The parameters σ_1 and σ_2 are used to balance the fidelity terms and other terms. It can be observed that the proposed model yields the maximum PSNR and requires the fewest iterations when $\sigma_1 = \sigma_2 = 1$.

3) Regularization Parameters λ and μ: The parameter μ is used to balance the cloud component regularization term. The proposed model yields the maximum PSNR when $\mu = 0.01$.

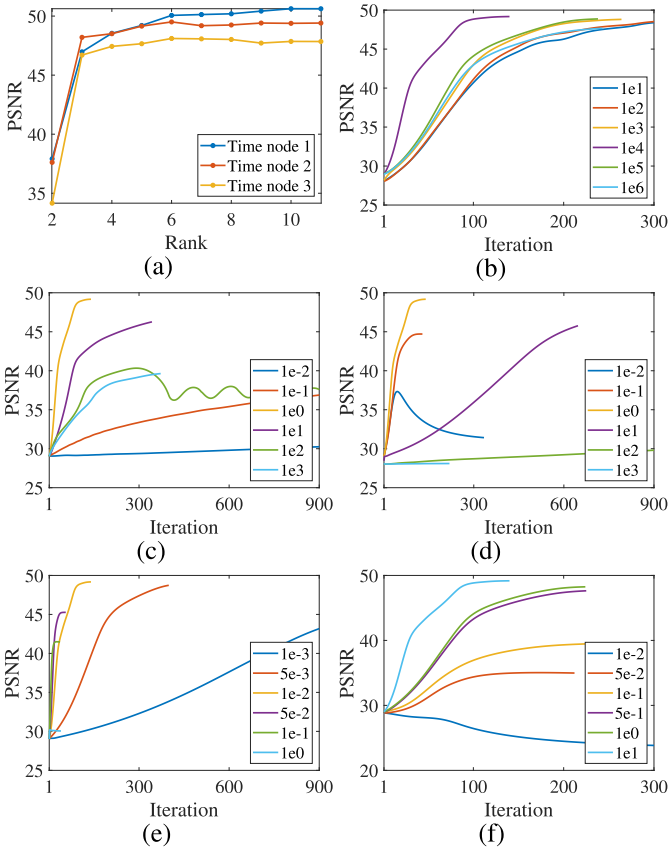


Fig. 6. Effects of parameters (a) r, (b) α, (c) σ_1, (d) σ_2, (e) μ, and (f) λ on model performance.

The parameter λ is used to balance the temporal smoothness regularization term. Larger λ value leads to better results. However, the effect of the proposed model is mostly converged when $\lambda \geq 10$, so we set $\lambda = 10$.

The penalty parameters α, σ_1, and σ_2 and the regularization terms parameters λ and μ selected in the experiments are applicable to both synthetic and real experiments. The proposed model requires a larger rank r to effectively handle larger scale MTRSIs in real data experiments.

D. Ablation Experiment

We conduct ablation experiments on the *China* dataset to quantitatively evaluate and verify the effectiveness of each module. The spatial factor-based group sparsity regularization, cloud component-based group sparsity regularization, and temporal smoothness regularization are successively discarded from model (6) until only the LRTF framework remains. Table V presents the cloud removal effect of the model after discarding the specific regularization terms, where TSR, GSRC, and GSRS represent temporal smooth regularization, group sparsity regularization on cloud component, and group sparsity regularization on spatial factor, respectively. It can be observed varying degrees of reduction in the cloud removal effect of the model as each regularization term is discarded. Thus, every module in model (6) is considered valid.

E. Convergence Analysis

In this section, we analyze the convergence of the developed PAM-based solving algorithm for proposed MTRSIs' cloud

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TABLE V
QUANTITATIVE RESULTS OF ABLATION EXPERIMENT ON THE CHINA DATASET

Module			PSNR	SSIM	SAM	CC
TSR	GSRC	GSRS				
-	-	-	33.0851	0.9540	0.4523	0.8820
✓	-	-	37.6412	0.9822	0.4045	0.9697
✓	✓	-	40.3712	0.9887	0.2916	0.9887
✓	✓	✓	**46.8112**	**0.9942**	**0.1482**	**0.9963**

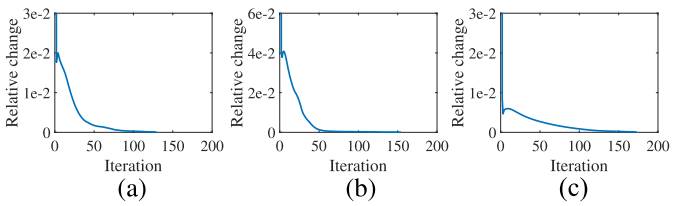


Fig. 7. Relative change values versus iteration numbers of the three datasets. (a) Brazil. (b) Italy. (c) China.

removal model numerically. Fig. 7 displays the relative change in reconstruction result at current iteration and previous iteration on the *Brazil*, *Italy*, and *China* datasets. It can be observed that the relative change values gradually decline and eventually approach zero as the iteration number increases, which verifies the convergence of the algorithm for MTRSIs' cloud removal model.

V. CONCLUSION

In this article, we propose a fast cloud removal method for MTRSIs using STCP. This method leverages group sparsity regularization within LRTF to explore the common structure between different spectral bands and time nodes of MTRSIs. It effectively avoids the high computational complexity issue of directly applying regularization on MTRSIs themselves. We develop an efficient algorithm based on PAM to solve the proposed nonconvex model and incorporate a thresholding algorithm of mask acquisition based on cloud and shadow component. In both synthetic and real experiments, the reconstruction results of the proposed STCP method demonstrate superior performance in terms of both overall and detailed aspects and exhibit significant improvement over other comparative methods in quantitative metrics. In the future, we will explore new tensor decomposition approaches to achieve more suitable and efficient factor regularization.

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